📘 File System

- Checking whether a file exists
- Listing Files in a Directory

📄 Overview

1. Checking Whether a File Exists

Conceptual Overview

Checking for file existence involves querying the operating system's file system to determine whether a given path refers to a valid object. On Windows, the **C++17's `<filesystem>` library** provides a standardized, cross-platform abstraction.

Prior to C++17, developers often:

- Tried opening the file using `<fstream>`
- Used Windows-specific APIs such as `GetFileAttributes`

Today, `<filesystem>` is preferred for correctness, portability, and clarity.

Using `std::filesystem::exists`

```
#include <filesystem>
#include <iostream>

namespace fs = std::filesystem;

int main() {
    fs::path filePath = "C:\\temp\\example.txt";

    if (fs::exists(filePath) && fs::is_regular_file(filePath)) {
        std::cout << "File exists.\n";
    } else {
        std::cout << "File does not exist.\n";
    }

    return 0;
}
```

2. Listing Files in a Directory

Conceptual Overview

Listing directory contents requires iterating over file system entries. In modern C++, provides iterator-based directory traversal that hides platform-specific details.

```
#include <filesystem>
#include <iostream>

namespace fs = std::filesystem;

int main() {
    fs::path directoryPath = "C:\\temp";

    if (!fs::exists(directoryPath) || !fs::is_directory(directoryPath)) {
        std::cout << "Invalid directory.\n";
        return 1;
    }

    for (const auto& entry : fs::directory_iterator(directoryPath)) {
        std::cout << entry.path() << '\n';
    }

    return 0;
}
```

Explanation

- fs::directory_iterator iterates over immediate directory entries.
- directory_iterator gives a std::filesystem::directory_entry object.
  - entry.path() returns the full path to the file or folder.
  - path() returns a std::filesystem::path object.
    - Useful members of the path object:
      - filename() - the name of the file without the path
      - extension() - the extension of the file

🔭 Markdown Viewer

How to view the markdown files in a browser...

- Markdown Viewer

🧠 Lecture Practices

Here are the lecture Practices...

- Day 10
- Day 11

🔍 Lecture Quizzes

Here are the lecture quizzes...

- Day 10
- Day 11

Weekly Topics

Here are the topics for the week...

- CSV
- File System
- Writing CSV
- Reading CSV
- Serializing
- Deserializing